

Identifying Obstacles to Equitable Flood Mitigation Funding

Russell Blessing^a, Julian Plough^b, Miyuki Hino^{a,b}, Antonia Sebastian^{a,b}, Helena Garcia^b

^a *University of North Carolina at Chapel Hill, Department of Earth, Marine and Environmental Sciences, Chapel Hill, USA,*

^b *University of North Carolina, Environment, Ecology and Energy Program, Chapel Hill, USA,*

Abstract

Federal flood mitigation funding in the United States flows to homeowners through a multi-stage application process administered by local governments. Identifying barriers to access has been constrained by data: prior studies observe only who receives funding, not who is eligible or who applies. We construct a novel parcel-scale dataset linking flood exposure across 78 events from 1996 to 2020 with FEMA Hazard Mitigation Assistance applications and funding outcomes for 3.14 million residential properties in eastern North Carolina. Of the 202,623 flood-exposed properties, about two-thirds reside in jurisdictions with at least one HMA application, yet only 5,746 (2.8%) apply, of which 3,432 (59.7%) are funded. Repeat-flooded properties apply at roughly twice the rate of once-flooded properties but are funded at similar rates, indicating that the application bottleneck applies even for properties for which mitigation would likely be cost-beneficial. Funded properties are concentrated in lower-value, less-white neighborhoods than the broader flood-exposed population, but within those neighborhoods they are typical rather than the cheapest homes, indicating that selection operates between neighborhoods rather than within them. Increasing application rates — particularly among repetitively flooded households — is the largest leverage point our data identify for expanding cost-effective mitigation investment.

Keywords: Flood mitigation, Climate resilience

1. Introduction

Flooding is the most costly natural hazard in the United States, and climate change is projected to intensify both its frequency and its financial burden

*Corresponding author

(for [Environmental Information](#), 2024). An estimated 41 million people live within the 1% flood hazard area, and the populations most exposed are disproportionately low-income, Black, and Native American ([Tate et al.](#), 2021). In response to mounting losses, federal investment in hazard mitigation assistance has grown substantially over the past decade, from \$748 million in 2015 to nearly \$2.8 billion in 2024, averaging over \$1.2 billion per year ([Federal Emergency Management Agency](#), 2024).¹ Yet several recent studies have suggested that this investment may not be reaching those who need it most. Nationwide analyses find that wealthier and whiter communities have secured more federal mitigation funding, while racially and economically marginalized communities have received comparatively less ([Mach et al.](#), 2019; [Elliott et al.](#), 2020; [Loughran and Elliott](#), 2022). Several mechanisms have been proposed to explain this pattern, including federal cost-benefit valuation that tends to favor higher-value properties ([Elliott et al.](#), 2020), variation in local government capacity to apply for and administer grants ([Mach et al.](#), 2019), and household-level barriers that shape mitigation behavior ([Bubeck et al.](#), 2012; [Binder et al.](#), 2023). Past studies, however, have relied on data on who receives funding, not who is eligible or who applies. As a result, it remains unclear where in the process inequities first emerge and which interventions would most effectively improve access.

Empirical evidence on each of these proposed mechanisms remains uneven. The most direct evidence comes from research on adjacent FEMA programs. In the Individuals and Households Program (IHP), which provides post-disaster assistance through a similar application-based structure, denials fall disproportionately on applicants in more socially vulnerable areas — those scoring higher on the Social Vulnerability Index (SoVI), a composite of demographic and socioeconomic Census indicators ([Drakes et al.](#) 2021). [Rivera, Jenkins, and Randolph](#) ([Rivera et al.](#), 2022) documented that following Hurricane Dolly in 2008, 50 to 71 percent of IHP applications from Rio Grande Valley colonias were rejected, largely because FEMA’s unclear definitions of “deferred maintenance” and “insufficient damages” disqualified households whose home damage could not be cleanly attributed to the disaster event. While these findings concern post-disaster assistance rather than pre-disaster mitigation, the underlying mechanism of application-stage barriers that disadvantage already-marginalized communities plausibly operates in HMA as well. FEMA itself has acknowledged that limited local capacity is a barrier to accessing federal mitigation resources. The agency convened a Place-Based Technical Assistance Task Force in 2023 to coordinate cross-agency support for communities that “have limited capacity to access and implement certain resources” ([Federal Emergency Management Agency](#), 2026), and within the Building Resilient Infrastructure and Communities (BRIC) program launched a Direct Technical Assistance initiative providing hands-on application support to under-resourced communities and tribal nations.

¹Federal share obligations across all FEMA Hazard Mitigation Assistance programs, computed from the OpenFEMA Hazard Mitigation Assistance Projects dataset, 2015–2024.

Here, we combine novel datasets spanning over two decades of flood exposure, grant applications, and funding allocations in North Carolina to examine the full flooded-to-funded pipeline (Figure 1). Flood exposure is derived from random forest models trained on address-level National Flood Insurance Program (NFIP) records, producing inundation footprints for 78 damaging events between 1996 and 2020 (García et al., 2025). We link these exposure records to property-level application and funding data for FEMA Hazard Mitigation Assistance (HMA) grant programs, provided by the North Carolina Department of Public Safety (NCDPS). Together, these datasets allow us to track individual properties across four stages (flooded, in an applying community, applied, and funded) and to examine how the composition of properties shifts in terms of flood exposure history, property value, and neighborhood racial demographics at each transition. This is the first analysis to follow properties from flood exposure through to mitigation funding outcome at the property level, making it possible to distinguish between households applying and not being selected and those that never enter the applicant pool at all.

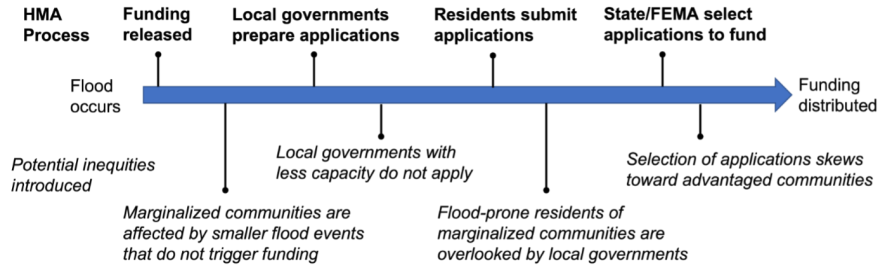


Figure 1: The application process for FEMA Hazard Mitigation Assistance funding presents different barriers to access at each stage. Diagram shows a simplified set of stages from flooded to funded (top), and the associated ways in which potentially inequitable outcomes could arise at each stage (bottom).

1.1. Access to federal funding for climate resilience

The federal government is the primary source of public assistance for disaster risk mitigation, post-disaster relief, and long-term recovery in the United States. Many federal agencies contribute to these efforts, including the US Army Corps of Engineers (which is responsible for large-scale flood-control infrastructure), the Department of Housing and Urban Development (which administers Community Development Block Grant Disaster Recovery funds), the National Oceanic and Atmospheric Administration, and the Environmental Protection Agency. We focus our analysis on the Federal Emergency Management Agency (FEMA), which leads disaster response and administers the nation’s largest hazard mitigation grants. These programs — delivered through FEMA’s Hazard Mitigation Assistance (HMA) framework — include the post-disaster Hazard Mitigation Grant Program (HMGP), which is triggered by presidential disaster declarations; Flood Mitigation Assistance (FMA), an annually appropriated

program targeting flood-insured properties; and Building Resilient Infrastructure and Communities (BRIC), a competitive pre-disaster program that replaced Pre-Disaster Mitigation (PDM) in 2020. FEMA made over \$1.6 billion available through BRIC and FMA alone in FY2024, with demand for these competitive funds regularly exceeding the available pool.

Concerns about equitable access to FEMA assistance in the aftermath of disasters have been extensively documented. Following presidential disaster declarations, FEMA’s Individuals and Households Program (IHP) provides up to approximately \$44,000 per household for temporary housing, property repair, unemployment assistance, and other immediate needs. Racial and income disparities in IHP access have been observed across a wide range of events, including the 1994 Northridge earthquake (Bolin and Stanford, 1998), Hurricane Katrina (Kamel, 2012), Hurricane Dolly in South Texas (Rivera et al., 2022), and the 2017 Thomas Fire (Méndez et al., 2020). More broadly, studies find that social vulnerability is inversely associated with access to federal disaster recovery funds even after accounting for the extent of damage (Emrich et al., 2022; Drakes et al., 2021; Wilson et al., 2021; Bento and Elliott, 2022).

Similar concerns have emerged around FEMA’s risk mitigation programs, which fund long-term adaptation rather than immediate recovery. Typical Hazard Mitigation Assistance (HMA) projects include home elevations, property buyouts, and community-scale infrastructure; since 2020, the BRIC program has expanded community-wide project funding. Among property-scale programs, buyouts have received the most scrutiny. In a buyout, homeowners sell flood-prone properties to local governments, which demolish structures and restore the land to open space. Initiated in 1985 and now totaling over 43,000 completed transactions nationally, the buyout program has shifted from its rural origins into an increasingly urban policy (Mach et al., 2019). Its geographic distribution is deeply uneven: states with the highest cumulative flood damages (Florida, Louisiana, Mississippi) rank among the lowest for buyout deployment, while activity concentrates in wealthier, denser counties with greater administrative capacity (Mach et al., 2019). Within those counties, bought-out properties are concentrated in lower-income and more racially diverse neighborhoods, a pattern Mach et al. (2019) describe as a “buyout paradox” in which government capacity governs access at the county scale while within-county selection follows a different logic (Elliott et al., 2020; Loughran and Elliott, 2022). Home elevation funding exhibits a different pattern, flowing disproportionately toward high-income and predominantly white communities (Frank, 2022).

1.2. Procedural challenges to accessing funding

While these structural barriers shape outcomes at a broad scale, the mechanisms that produce inequity operate at the level of specific program rules and local implementation practices. The HMA application process is lengthy and multi-layered. After federal funding is released, state, tribal, and local governments identify eligible properties, assist homeowners in compiling required documentation (including cost-benefit analyses, insurance records, and title documentation),

and submit project proposals. Local governments (“sub-applicants”) forward proposals to the state (“applicant”), which screens projects before submitting to FEMA for final review. For HMGP-funded buyouts, the full process from disaster declaration to project closeout averages nearly six years (Mach et al., 2019). The federal government typically covers 75% of project costs, with states, local governments, and in some cases homeowners responsible for the remaining 25% non-federal match. Higher federal shares apply in specific circumstances — up to 90% for small impoverished communities across programs, and, under FMA, up to 90% for Repetitive Loss and 100% for Severe Repetitive Loss properties.

Inequities can arise at every stage. At the local government level, administrative capacity is the primary bottleneck: governments with dedicated staff, prior grant experience, and stronger institutional networks are far more effective at developing and submitting competitive proposals (Mach et al., 2019; Junod et al., 2021). At the community and household level, outreach practices vary widely; some local governments proactively contact eligible residents and guide them through the application process, while others do little or nothing, effectively limiting access to only those homeowners already aware of the program. Providing the 25% non-federal cost share can also be a barrier. In some states, including North Carolina, the state government covers the matching share for HMGP buyout projects under disaster declarations (North Carolina Department of Public Safety, 2026). Where the state does not, the funding must come from local government budgets or individual homeowners, which can significantly constrain participation among cash-strapped jurisdictions and lower-income households.

At the household level, procedural requirements such as documentation, inspections, and legal title verification can impose substantial compliance burdens. While these requirements apply uniformly to applicants, the practical difficulty of meeting them is likely to vary systematically with household resources, mirroring documented patterns in IHP access (Rivera et al., 2022; Raker, 2023). Finally, federal cost-benefit analysis (CBA) requirements may further filter the pool. CBA evaluates projects on a benefit-to-cost ratio in which avoided losses scale with property replacement value, leading to concerns that applications from lower-value properties could face systematic disadvantages in the benefit-to-cost ratio (Tate et al., 2021; Junod et al., 2021; Miller, 2023). However, project costs also scale with property characteristics, and CBA exemptions exist for substantially damaged structures below specified cost thresholds, so the net distributional effect of CBA on funding outcomes remains an open empirical question.

2. Results

We track 3.14 million residential parcels across 78 counties in eastern North Carolina through four sequential stages of the federal HMA pipeline: flooded (intersection with any of 78 modeled flood-event footprints from 1996 through 2020), in an applying community (jurisdiction with at least one HMA application

submitted during the study period), applied (parcel itself named on an HMA application), and funded (application ultimately awarded). Flood-exposure data come from a random-forest model of NFIP claims and policies-in-force (Garcia et al., 2025), application and funding records from the North Carolina Department of Public Safety, and neighborhood demographics from the 2013 American Community Survey at the block-group scale. Property values come from the CoreLogic database and are expressed as percentile ranks across the study-area sample. Detailed methods are described in the Materials and Methods section.

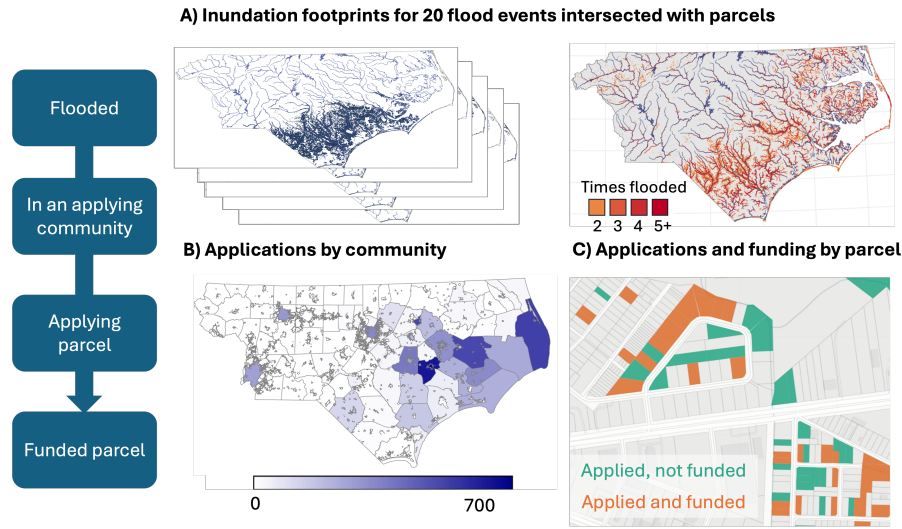


Figure 2: **Spatial distribution of properties across the flooded-to-funded pipeline.** Left: schematic of the four sequential pipeline stages tracked in our analysis. Top-left: flooded parcels ($n = 202,623$) across the 78-county study area, with stacked footprints from 78 modeled flood events between 1996 and 2020. Top-right: repetitive residential flooding, with cells colored by the number of modeled flood events affecting each location (2, 3, 4, or 5+ events). Bottom-left: jurisdictions classified as applying communities (shaded by share of eligible parcels named on at least one HMA application). Bottom-right: representative neighborhood showing individual parcels named on HMA applications, distinguished by funding outcome (applied but not funded vs. applied and funded).

While most flooded properties are in communities that have applied for funding, very few properties submit applications themselves (Figure 3). Among all flooded properties (Figure 3, left panel), 65.6% are in jurisdictions where local government assisted with the submission of at least one HMA application. Only 2.8% of flooded properties are themselves named on an application, and 59.7% of those applications are funded (only 1.7% of all flooded properties).

Approximately 32.8% of flooded properties were exposed to more than one of the 78 modeled events (Figure 3, right panel). Repeat-flooded properties are more likely to be located in jurisdictions with at least one HMA application (81.1% versus 65.6% among all flooded properties) and themselves apply at

roughly twice the rate of once-flooded properties (5.6% versus 2.8%). However, the share of applications that ultimately receive funding is essentially the same across the two groups (58.5% versus 59.7%), indicating that while greater flood exposure increases the likelihood of applying, it does not appear to influence selection outcomes among those who do apply.

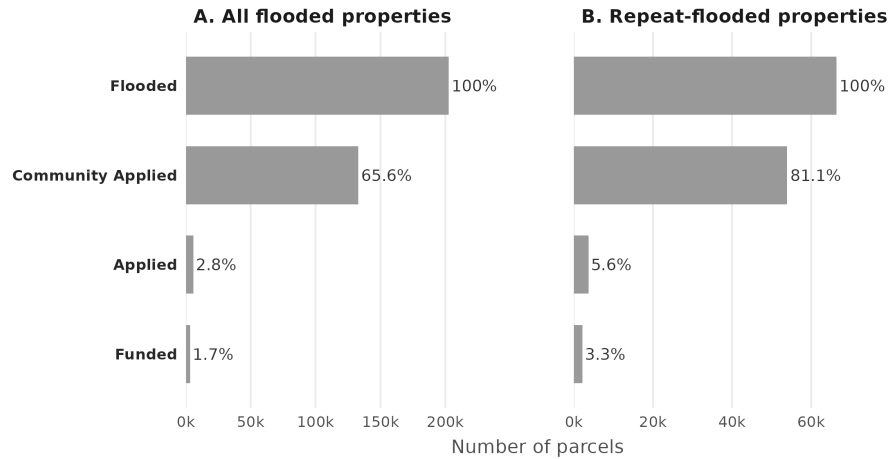


Figure 3: **The application stage is the largest attrition point in the HMA pipeline: fewer than 3% of flooded properties ever submit an application.** Bars show the number of 1–4 family residential parcels at each of four pipeline stages: flooded in any of the 78 modeled events; in an applying community (jurisdiction with at least one HMA application during the study period); itself named on an HMA application; and ultimately funded. Panel A: all flood-exposed parcels ($n = 202,623$). Panel B: parcels exposed to flooding in more than one event ($n = 66,443$). Repeat-flooded properties apply at roughly twice the rate of once-flooded properties but are funded at similar rates.

Property values shift substantially across pipeline stages, both relative to the broader study area and within each property’s own block group (Figure 4). Properties that flooded or that reside in communities with at least one HMA application have median values near the 51th–53th percentile of the study-area sample (Panel A). The median value for properties that apply for funding drops to the 29th percentile, and the median for those that are funded is even more affordable at the 23rd percentile. Panel B re-ranks each property against other parcels in its own block group, with all stages clustering near the 60th percentile (Flooded: 61st, Funded: 57th). The much larger spread in Panel A, where flooded properties are within the 51st percentile and funded properties at the 23rd, indicates that the property-value shift across pipeline stages is driven by selection between neighborhoods rather than within them. Funded properties are typical homes in lower-value neighborhoods, not the lowest-value homes in any given neighborhood.

Properties applying for and receiving mitigation funding are located in block groups with lower shares of non-Hispanic White residents and correspondingly

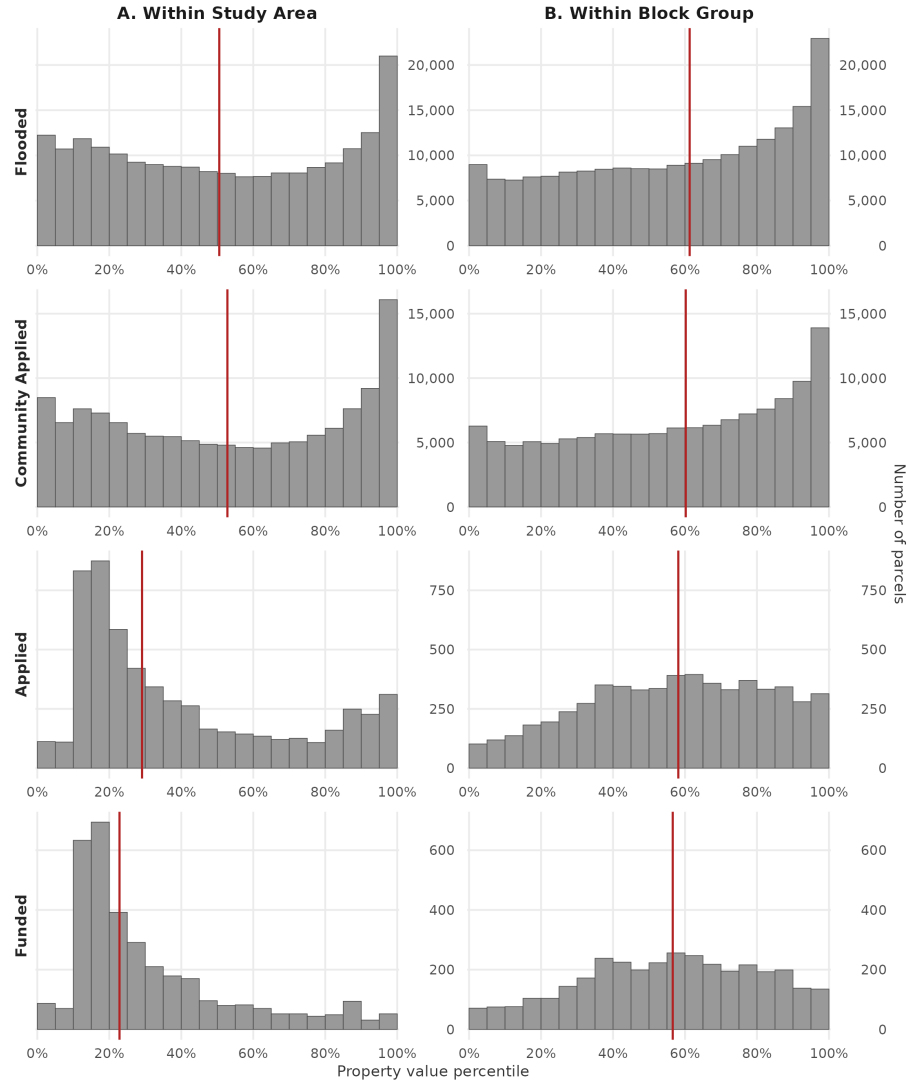


Figure 4: **Funded properties average lower property values than the broader flood-exposed population, but are typical homes for their neighborhoods.** (A) Distribution of property values for parcels reaching each stage of the flooded-to-funded pipeline, expressed as percentile ranks relative to the entire study-area sample. (B) The same distributions, expressed as percentile ranks within each parcel's block group. Property values are from CoreLogic (2022 dollars); red vertical lines mark stage medians.

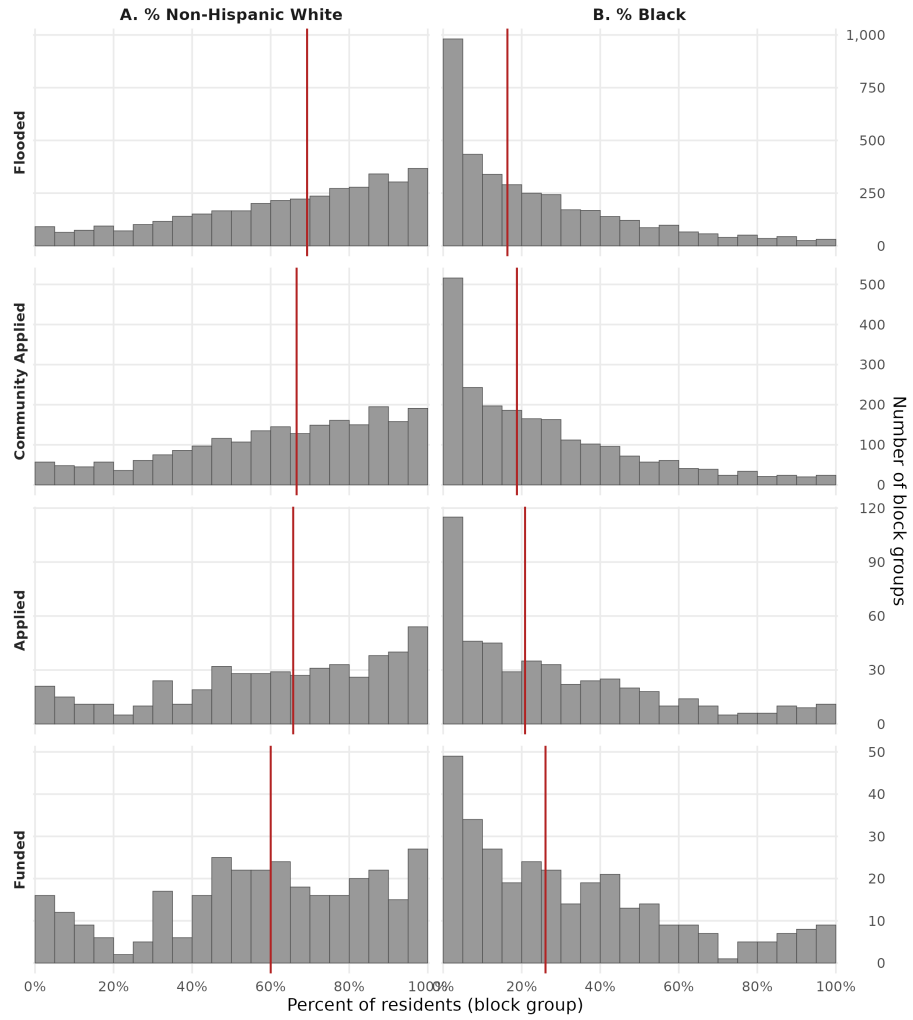


Figure 5: **Funded properties are more likely to be in less-white, more-Black neighborhoods than the broader flood-exposed population.** (A) Percentage of non-Hispanic White residents. (B) Percentage of Black residents. One observation per block group containing 1 residential parcel reaching the stage; demographic data from the 2013 ACS 5-year estimates. Red vertical lines mark stage medians.

higher shares of Black residents than block groups at preceding pipeline stages (Figure 5). Two-sample Kolmogorov-Smirnov tests confirm that the distributions of block-group racial composition at the funded stage differ significantly from those at the flooded stage (NH White: $D = 0.131$, $p < 0.001$; Black: $D = 0.151$, $p < 0.001$). Median NH White share declines from 69.3% at the flooded stage to 60.0% at funded properties, with a near-symmetric rise in Black share from 16.4% to 26.1%.

The racial composition of funded parcels differs substantially by mitigation type (Figure 6), which contributes to the funded-stage shift documented in Figure 5. Among funded parcels with a classified project type, buyouts outnumber elevations by approximately 4:1 (1,879 buyouts vs 415 elevations). Buyouts are located in block groups with substantially higher shares of Black residents than elevations (median 47% versus 35.7%; Kolmogorov-Smirnov $D = 0.366$, $p < 0.001$), while property-value differences between mitigation types are more modest (median 20th versus 25.7th percentile; $D = 0.141$, $p < 0.001$). The pattern holds in the repeat-flooded subset ($D = 0.32$ for racial composition), indicating that the buyout-elevation divergence is not driven by event-specific dynamics. Because roughly one-third of funded parcels have project types we could not classify, we cannot fully attribute the Figure 5 shift to either mechanism, but the combination of buyouts' higher prevalence and their concentration in less-white neighborhoods suggests they are a meaningful contributor.

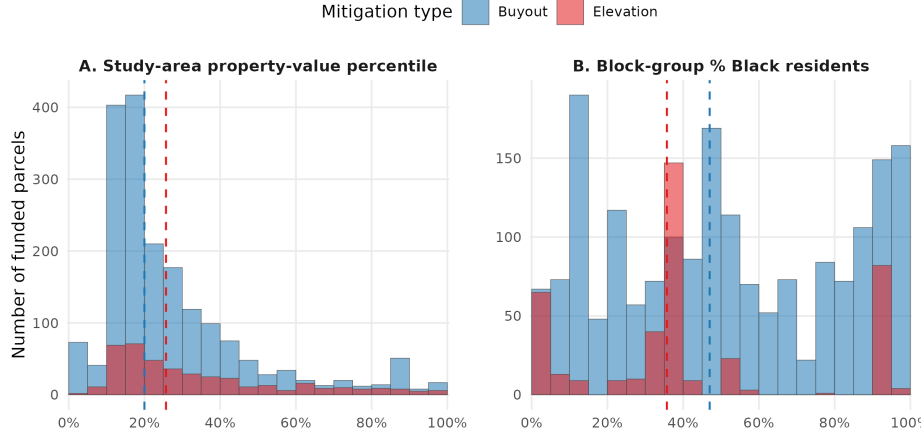


Figure 6: **The racial composition shift at the funded stage is driven primarily by buyouts, while property-value differences between mitigation types are modest.** Distribution of (A) study-area property-value percentile rank and (B) block-group share of Black residents for funded parcels classified as buyouts (Acquisition, blue) and elevations (Elevation, red). Dashed vertical lines mark each group's median. Funded parcels without a recorded project-type classification (approximately one-third of the funded set) are excluded. Statistical comparison: property value $D = 0.141$, % Black $D = 0.366$ (both $p < 0.001$).

3. Discussion

Our analysis identifies the application stage as the critical pinch point in the flooded-to-funded HMA pipeline. Of all flooded residential properties in eastern North Carolina, only 2.8% apply for funding, and of those, 59.7% receive an award. By tracking the full population of flooded properties rather than only those that received funding, we identify a substantial population of households who would benefit from mitigation assistance but do not appear in the applicant pool. Many of these properties have flooded repeatedly, suggesting that mitigation would likely be cost-beneficial and that they would meet the benefit-cost thresholds required for federal funding. This unrealized demand, to our knowledge not previously quantified at the property scale, suggests that interventions targeting the application stage may shift the composition of who ultimately receives funding more than further refinement of the selection criteria applied to existing applicants. Because HMA-funded mitigations have repeatedly been found to be cost-effective at the project level (Council, 2017), expanding the applicant and funding pool among flooded households would translate to more cost-effective mitigation in aggregate, not merely a more competitive selection process among existing applicants.

Among the small fraction of properties that do receive funding, the composition shifts toward lower-value, less-white neighborhoods, indicating that the selection process among applicants is not itself driving inequitable funding outcomes. Funded properties are within the 23rd percentile of the study-area property-value distribution (compared to the 51st at the flooded stage) and in block groups with a median non-Hispanic white share of 60% (compared to 69.3%) and Black share of 26.1% (compared to 16.4%). The shift occurs primarily between neighborhoods rather than within them: within-block-group percentiles for funded properties differ only modestly from those at the flooded stage. This pattern aligns with the neighborhood-scale evidence in prior work (Mach et al., 2019; Elliott et al., 2020), which finds that within whiter, wealthier counties, funding concentrates in lower-value, less-white neighborhoods. Because we do not conduct an analogous county-level comparison, our results speak to this within-county scale rather than the county-scale pattern those studies also document. It likely reflects the strong correlation between race and property value in eastern North Carolina, where less-white neighborhoods tend to also be lower-value (Figure 9).

Repeat-flooded properties apply at roughly twice the rate of once-flooded properties but are funded at similar rates. This pattern suggests that flood-exposure history influences application behavior more than selection outcomes, and points to an underused targeting lever for HMA prioritization. HMGP, which dominates HMA funding in our study area, becomes available event-by-event following PDDs and weights damage from a single event heavily in the cost-benefit calculation. Mitigation programs that gave greater weight to a property’s cumulative flood history, rather than only the severity of a single damaging event, would more closely align selection with the population our data

identifies as most likely to be affected again.

Decomposing the funded set by mitigation type reveals that the racial composition shift at the funded stage is driven primarily by buyout activity rather than elevation activity (Figure 6). Buyouts dominate the funded record (approximately 4:1 over elevations among classified parcels) and concentrate in block groups with substantially higher shares of Black residents than elevations (KS $D = 0.366$, $p < 0.001$). Property-value differences between mitigation types are more modest ($D = 0.141$), suggesting that the property-value shift across the pipeline reflects a neighborhood-scale selection process affecting both mitigation types similarly, while the racial composition shift reflects a buyout-specific dynamic. This pattern extends prior county-scale findings of demographic differences between buyout and elevation targeting (Mach et al., 2019; Elliott et al., 2020; Frank, 2022) to the parcel scale within neighborhoods, and persists in the repeat-flooded subset ($D = 0.32$), suggesting it reflects structural differences in how buyouts and elevations are sited rather than event-specific dynamics.

Our cross-sectional data also let us rule out several candidate mechanisms for the equity patterns we observe. If the federal CBA requirement were primarily driving selection toward higher-value properties, we would expect funded properties to be concentrated at higher property values; instead, we see the opposite, especially given that we observe many lower-value properties applying. Because CBA is a federal rule applied uniformly across states, this pattern should generalize beyond North Carolina. The high community participation rate in our study area (65.6% of flooded properties are in jurisdictions with at least one HMA application) similarly argues against wholesale community-level non-participation as the binding constraint, though this likely differs in states without state-level cost-share absorption like North Carolina’s. We also see no evidence of applicant pools dominated by high-value properties, which would be consistent with local governments selectively recruiting wealthier households for application support, though this pattern may be specific to North Carolina and could differ in states with different local capacity or outreach practices. Together, these patterns suggest that the concentration of funding in lower-value, less-white communities is determined primarily at the application stage — where the median shift toward these communities is largest — rather than by selection within the applicant pool, where the applied-to-funded shift is comparatively modest. In other words, the pattern appears to be a function of who applies, and who is solicited to apply, more than which applications are chosen. Disentangling whether this reflects differential outreach, programmatic preferences for buyout siting, or other mechanisms is an important direction for future work.

Several methodological limitations apply. First, our flood-exposure data come from a reconstruction of 78 flood events in eastern North Carolina between 1996 and 2020 (Garcia et al., 2025). The underlying models are conservative and likely underestimate the true population of flood-exposed properties. A sensitivity analysis dropping the seven events Garcia et al. flag for unusually high modeled-to-observed damage ratios produces materially identical headline findings: the

funded population shifts by only 2.3% and median demographic compositions move by less than half a percentage point (Supplementary Materials). Second, our data on the timing of applications and mitigations is incomplete, so we do not distinguish between floods that occur before or after a property is funded for mitigation. This is particularly relevant for our repeat-flood analysis: properties that were elevated or bought out before a subsequent modeled flood event may appear in our data as repeat-flooded even though the structure may no longer have been exposed or may have been physically removed. Because these cases require a property to have already been funded for mitigation, they should be concentrated in the funded stage and are unlikely to be numerous; they would, if anything, modestly overstate the funding rate among repeat-flooded properties rather than change our central conclusion that repeat-flooded properties apply at higher rates but are funded at similar rates to once-flooded properties. Third, approximately one third of funded mitigation records in our data lack a project-type classification, so our buyout-versus-elevation comparison (Results section, Figure 6) is restricted to the classified subset. The unclassified records may not be randomly distributed across mitigation types, and any systematic gap (for example, if smaller or older projects disproportionately lack documentation) could affect the magnitude of the buyout-elevation contrast we report. Fourth, our block-group demographic data come from the 2013 American Community Survey, which falls near the midpoint of the study window but does not capture demographic changes over the full 1996–2020 period. Finally, while we document more of the application process than past studies, there are still many aspects that we do not observe, such as how local governments approach (or fail to approach) flooded households about funding opportunities, procedural obstacles that may deter applications from interested households, and sources of non-governmental assistance.

A broader interpretive caveat concerns what our findings imply for affected households. Federal mitigation funding flows disproportionately to lower-value, less-white block groups relative to the broader flood-exposed population, a pattern consistent with mitigation reaching historically underserved communities in North Carolina. Whether participation actually serves these households, however, requires post-mitigation outcome data we do not have; this is particularly relevant for buyouts, where prior research has questioned whether participation reflects informed preference or limited alternatives (Mach et al., 2019). Within block groups, funded properties are typical of the local housing stock rather than the cheapest, suggesting that within-community project selection does not systematically filter out lower-value homes. The broader compositional shift we observe is determined largely by upstream pipeline stages such as community participation, application rates, and the spatial distribution of flood exposure itself, rather than by within-community project selection. Procedural neutrality at any one step is not sufficient to establish that the cumulative distribution of funding is substantively equitable.

3.1. Conclusions

This study follows individual residential properties from flood exposure through mitigation funding outcome at each significant stage, including the previously unobserved application stage. The largest falloff in the pipeline occurs at the application step, and properties that reach subsequent stages tend to be lower-value and concentrated in less-white neighborhoods, primarily through between-neighborhood selection rather than within-neighborhood selection. These results identify the application stage as a leverage point for expanding access to federal mitigation funding, one that prior work focused on funded outcomes alone has been unable to observe.

These findings arrive amid significant federal restructuring of hazard mitigation policy. The FEMA Review Council’s May 2026 final report recommended replacing HMGP with a state-managed Refined Risk Reduction Program that would prioritize repetitively flooded properties and acknowledged administrative burdens that may discourage applications from under-resourced communities ([President’s Council to Assess the Federal Emergency Management Agency, 2026](#)). The application-stage attrition we document is consistent with those concerns. Reforms targeting funding-side changes such as disbursement speed, federal priority-setting, or program structure will require complementary attention to application-stage barriers if they are to translate into broader access. Future research could extend this approach to community-scale infrastructure investments such as stormwater system upgrades and BRIC community projects, which may also reduce household-level flood risk through complementary pathways.

4. Methods

We assemble over 20 years of data on flood exposure, applications for state and federal funding, and funding allocations. Flood exposure data are generated using a random forest model trained on address-level records of National Flood Insurance Program (NFIP) policies-in-force and claims filed from January 1996 through September 2020, obtained from FEMA Region IV ([García et al., 2025](#)). The model domain encompasses the entirety of the Neuse–Pamlico and Cape Fear watersheds and portions of the Pee Dee and Chowan–Roanoke watersheds that drain through eastern North Carolina, overlapping with 78 of North Carolina’s 100 counties. This measure indicates whether a parcel intersected the modeled footprint of an event but does not capture the severity of any individual exposure; some properties classified as flooded may have experienced relatively minor inundation. We link these flood exposure data to parcel geometries from the North Carolina OneMap database, attribute records from the CoreLogic property database, application and funding records from the North Carolina Department of Public Safety (NCDPS), and block-group demographic data from the American Community Survey (ACS).

We use this combined dataset to track residential properties through four sequential stages of the flood mitigation funding pipeline (Figure 2): whether

a property has flooded, whether its local government has applied for funding, whether the property itself is named on an application, and whether that application was funded.

4.1. Unit of analysis

The base unit of analysis is the residential parcel. Parcel geometries come from the North Carolina OneMap database, which aggregates county tax assessor records into a statewide spatial layer, and each parcel is labeled with property-level information (land use code, structure characteristics, assessor value) from the CoreLogic property database. Where parcel records and CoreLogic records share a common identifier we match directly; the remaining parcels are matched spatially, with address-string similarity used to break ties when more than one CoreLogic record is closest. We retain 1–4 family residential parcels and exclude commercial, industrial, agricultural, and most government-owned land uses. A small number of residential parcels are recorded in CoreLogic as currently vacant or government-owned. Because completed buyouts demolish the structure and transfer the parcel to public ownership, these post-buyout parcels appear in the present-day data with property values that reflect their post-mitigation rather than pre-mitigation state. To recover a pre-mitigation value for parcels that we have reason to believe were previously residential, we impute the property value as the mean assessor value of the five nearest non-vacant residential parcels. Parcels that are vacant for reasons unrelated to mitigation are excluded from the property-value analysis. The final analytic sample comprises 3,137,520 parcels.

4.2. Flooded-to-funded pipeline stages

Flood exposure is constructed at the parcel level using the inundation footprints generated by Garcia et al. (2025), which resolve individual flood events on a 30 m $\times$ 30 m grid. We retain the 78 events with the largest claim counts in the study area, which together account for 95% of all NFIP claims filed over the period (Garcia et al., 2025). A parcel is classified as flooded if its geometry intersects the footprint of at least one of these 78 events. This property-resolved flood history serves as the reference set against which subsequent pipeline stages are measured, enabling the first parcel-level linkage between flood exposure and federal mitigation outcomes in the region.

Application and funding records cover two FEMA Hazard Mitigation Assistance (HMA) programs administered by NCDPS: the Hazard Mitigation Grant Program (HMGP), which becomes available following presidential disaster declarations (PDDs) issued event-by-event and county-by-county; and Flood Mitigation Assistance (FMA), which depends on a property’s NFIP claim and policy history rather than on a PDD. FMA encompasses the repetitive-loss pathways previously administered under the standalone Repetitive Flood Claims (RFC) and Severe Repetitive Loss (SRL) programs before their consolidation into FMA in FY2014. Of the 78 events in our flood-exposure dataset, 18 triggered PDDs that included at least one North Carolina county; these constitute the universe of disaster declarations from which HMGP funding could flow during our study period.

NCDPS provided address-level records of every project submitted for HMA funding under HMGP and FMA, along with the subset of those projects that were ultimately funded. Application records ($n = 23,232$) and funded mitigation records ($n = 8,801$) span 1996 through 2018, encompassing major North Carolina disaster events including Hurricanes Fran (1996), Floyd (1999), Isabel (2003), Irene (2011), Matthew (2016), and Florence (2018). Hurricane Floyd dominates the funded record, accounting for 58% of mitigations with a recorded disaster year.

The HMA application process is administered through local governments. Households cannot apply directly to FEMA: instead, a county or municipal government compiles project documentation (including the cost-benefit analysis, NFIP claim records, and deed and title records) and submits applications on behalf of interested homeowners, often with state-level coordination from NCDPS. Local governments differ substantially in their willingness and capacity to do this work (Mach et al., 2019; Junod et al., 2021). To capture this community-level filter, we assign every parcel to its containing municipality, or to its containing county where no municipality exists, and classify each jurisdiction by whether it submitted at least one HMA application during the study period. Within applying jurisdictions, we then geocode application and funding records back to OneMap parcels and classify each property as never applied, applied but not funded, or applied and funded.

4.3. Pipeline analysis

Property values are taken from the most recent CoreLogic vintage and used at the parcel level. We express each property’s value as a percentile rank across all 1–4 family residential parcels in the study area, using relative rather than absolute property value as our primary property-value measure throughout the analysis. To complement the broad ranking, we also compute each property’s value as a percentile rank within its own census block group. This within-neighborhood percentile lets us test whether selection toward lower-value properties occurs within neighborhoods or primarily reflects selection toward lower-value neighborhoods.

Neighborhood-level race and income data are obtained at the census block-group scale and joined to each parcel based on its containing block group. We use the 2013 vintage of the American Community Survey because it falls near the midpoint of our 1996–2020 flood-exposure window. The 18 disaster declarations represented in our study area span this period, with the earliest being Hurricane Fran (1996) and the most recent being Hurricane Florence (2018). Block-group variables include the share of residents identifying as non-Hispanic white, the share identifying as non-Hispanic Black, and median household income. We aggregate parcel-level results to the block group when reporting demographic-composition shifts across pipeline stages.

The four-stage pipeline is a sequence of nested subsets: each stage is, by construction, a subset of the previous one. We characterize each stage in two ways. First, we count the number of parcels that survive into the stage and

report stage-to-stage attrition rates for both once-flooded parcels and parcels that flooded two or more times (i.e., repeatedly flooded parcels). Our flood-exposure measure captures only the number of events a parcel intersected, not the severity (such as depth, duration, or damage) of any individual exposure. Second, we describe the distribution of property values (study area wide percentile rank) and block-group demographics at each stage, to identify where in the pipeline shifts in wealth and racial composition emerge.

This cross-sectional design intentionally treats the pipeline as a series of selection filters. By observing applications, funding, and flood exposure at the parcel level for every residential property in the study area, we identify the transitions at which inequities emerge, rather than inferring them from funded properties alone.

4.4. Mitigation type classification

To examine whether the wealth and racial-composition shifts we observe at the funded stage differ between buyouts and elevations, we classify each funded mitigation record by mitigation type. NCDPS records distinguish acquisitions (in which a property is purchased and the structure demolished, requiring household relocation) from elevations (in which the existing structure is raised above the flood level, preserving in-place residence). We classify parcels with at least one acquisition record as buyouts, parcels with at least one elevation record (but no acquisition) as elevations, and the remainder — including Utility Infrastructure, Quick Connect, and records with no mitigation type specified — as Other/Unknown. Approximately one third of funded parcels fall into the Other/Unknown category, and the comparison below is restricted to the classified subset of 2,294 parcels.

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References

- Bento, A., Elliott, J.R., 2022. The Racially Unequal Impacts of Disasters and Federal Recovery Assistance on Local Self-Employment Rates. *Social Currents* 9, 424–444. doi:[10.1177/23294965211028841](https://doi.org/10.1177/23294965211028841).
- Binder, S.B., Baker, C.K., Ritchie, L.A., Barile, J.P., Greer, A., 2023. “Upheaval”: Unpacking the dynamic balance between place attachment and social capital in disaster recovery. *American Journal of Community Psychology* 72, 378–394. doi:[10.1002/ajcp.12697](https://doi.org/10.1002/ajcp.12697).
- Bolin, R., Stanford, L., 1998. The Northridge Earthquake: Community-based Approaches to Unmet Recovery Needs. *Disasters* 22, 21–38. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/1467-7717.00073>, doi:[10.1111/1467-7717.00073](https://doi.org/10.1111/1467-7717.00073). _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1467-7717.00073>.
- Bubeck, P., Botzen, W.J.W., Aerts, J.C., 2012. A review of risk perceptions and other factors that influence flood mitigation behavior. *Risk Analysis: An International Journal* 32, 1481–1495.
- Council, M.M., 2017. Natural Hazard Mitigation Saves: 2017 Interim Report. Technical Report. National Institute of Building Sciences. Washington, DC. URL: <https://www.nibs.org/page/mitigationsaves>.
- Drakes, O., Tate, E., Rainey, J., Brody, S., 2021. Social Vulnerability and Short-Term Disaster Assistance in the United States. *International Journal of Disaster Risk Reduction* 53, 102010. doi:[10.1016/j.ijdr.2020.102010](https://doi.org/10.1016/j.ijdr.2020.102010).
- Elliott, J.R., Brown, P.L., Loughran, K., 2020. Racial Inequities in the Federal Buyout of Flood-Prone Homes: A Nationwide Assessment of Environmental Risk. *Socius: Sociological Research for a Dynamic World* 6, 1–13. doi:[10.1177/2378023120905439](https://doi.org/10.1177/2378023120905439).
- Emrich, C.T., Aksha, S.K., Zhou, Y., 2022. Assessing Distributive Inequities in FEMA’s Disaster Recovery Assistance Fund Allocation. *International Journal of Disaster Risk Reduction* 74, 102855. doi:[10.1016/j.ijdr.2022.102855](https://doi.org/10.1016/j.ijdr.2022.102855).
- for Environmental Information, N.N.C., 2024. U.S. Billion-Dollar Weather and Climate Disasters. URL: <https://www.ncei.noaa.gov/access/billions/>, doi:[10.25921/stkw-7w73](https://doi.org/10.25921/stkw-7w73).
- Federal Emergency Management Agency, 2024. Hazard Mitigation Assistance Projects (V4). <https://www.fema.gov/openfema-data-page/hazard-mitigation-assistance-projects-v4>. OpenFEMA dataset; accessed 2026-06-25.
- Federal Emergency Management Agency, 2026. Place-Based Technical Assistance Task Force. <https://www.fema.gov/emergency-managers/national-preparedness/frameworks/mitigation/mitflg/place-based-task-force>. Accessed: 2026-06-25.

- Frank, T., 2022. FEMA home elevation funds could violate civil rights law. E&E News by POLITICO URL: <https://www.eenews.net/articles/fema-home-elevation-funds-could-violate-civil-rights-law/>.
- Garcia, H.M., Sebastian, A., Fitzmaurice, K.P., Hino, M., Collins, E.L., Characklis, G.W., 2025. Reconstructing Repetitive Flood Exposure Across 78 Events From 1996 to 2020 in North Carolina, USA. *Earth's Future* 13, e2025EF006026. URL: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025EF006026>, doi:10.1029/2025EF006026.
- Junod, A.N., Martín, C., Marx, R., Rogin, A., 2021. Equitable Investments in Resilience: A Review of Benefit-Cost Analysis in Federal Flood Mitigation Infrastructure. Technical Report. Urban Institute, Metropolitan Housing and Communities Policy Center. URL: <https://www.urban.org/research/publication/equitable-investments-resilience>.
- Kamel, N., 2012. Social Marginalisation, Federal Assistance and Repopulation Patterns in the New Orleans Metropolitan Area following Hurricane Katrina. *Urban Studies* 49, 3211–3231. doi:10.1177/0042098011433490.
- Loughran, K., Elliott, J.R., 2022. Unequal Retreats: How Racial Segregation Shapes Climate Adaptation. *Housing Policy Debate* 32, 171–189. doi:10.1080/10511482.2021.1931928.
- Mach, K.J., Kraan, C.M., Hino, M., Siders, A.R., Johnston, E.M., Field, C.B., 2019. Managed Retreat Through Voluntary Buyouts of Flood-Prone Properties. *Science Advances* 5, eaax8995. doi:10.1126/sciadv.aax8995.
- Miller, B.M., 2023. The Cost of Cost-Effectiveness: Expanding Equity in Federal Emergency Management Agency Hazard Mitigation Grants. Technical Report. RAND Corporation. URL: https://www.rand.org/pubs/research_reports/RR2171-1.html.
- Méndez, M., Flores-Haro, G., Zucker, L., 2020. The (in)visible victims of disaster: Understanding the vulnerability of undocumented Latino/a and indigenous immigrants. *Geoforum* 116, 50–62. URL: <https://www.sciencedirect.com/science/article/pii/S0016718520301925>, doi:10.1016/j.geoforum.2020.07.007.
- North Carolina Department of Public Safety, 2026. Disaster hazard mitigation grants. URL: <https://www.ncdps.gov/HMGP>. accessed June 2026.
- President's Council to Assess the Federal Emergency Management Agency, 2026. Final report: The President's Council to Assess the Federal Emergency Management Agency. URL: https://www.dhs.gov/sites/default/files/2026-05/26_0507_fema_review_council_final_report.pdf.
- Raker, E.J., 2023. Stratifying Disaster: State Aid, Institutional Processes, and Inequality in American Communities. *Social Forces* 102, 430–453. doi:10.1093/sf/soad050.

- Rivera, D.Z., Jenkins, B., Randolph, R., 2022. Procedural Vulnerability and Its Effects on Equitable Post-Disaster Recovery in Low-Income Communities. *Journal of the American Planning Association* 88, 220–231. URL: <https://doi.org/10.1080/01944363.2021.1929417>, doi:10.1080/01944363.2021.1929417. _eprint: <https://doi.org/10.1080/01944363.2021.1929417>.
- Tate, E., Rahman, M.A., Emrich, C.T., 2021. Flood Exposure and Social Vulnerability in the United States. *Natural Hazards* 106, 435–457. doi:10.1007/s11069-020-04470-2.
- Wilson, B., Tate, E., Emrich, C.T., 2021. Flood Recovery Outcomes and Disaster Assistance Barriers for Vulnerable Populations. *Frontiers in Water* 3, 752307. doi:10.3389/frwa.2021.752307.

Supplementary Materials

The supplementary materials include additional figures and analyses that support the main-text results. Section S1 presents block-group economic and racial composition across the HMA pipeline. Section S2 documents the correlation between race and economic indicators in the study area. Section S3 provides the full buyout-versus-elevation comparison including the repeat-flooded subset. Section S4 reports a sensitivity analysis that re-runs the full pipeline excluding seven high-uncertainty flood events.

S1. Block-group economic distributions across the HMA pipeline

S2. Race and economic correlation in the study area

S3. Buyouts versus elevations across all and repeat-flooded funded parcels

The main-text comparison of buyouts and elevations (Figure 6 in main text) restricts to the “all funded” subset. The figures below replicate that comparison while also showing the repeat-flooded subset (Panel B), to verify that the buyout-elevation divergence is not driven by event-specific dynamics.

S4. Sensitivity to high-uncertainty events

Garcia et al. (2025) flag seven of the 78 events as having unusually high ratios of modeled-flooded buildings to NFIP claims (more than 20 buildings per claim), indicating greater uncertainty in those event reconstructions. We rebuilt the full flooded-to-funded pipeline excluding these seven events to verify that our headline findings do not depend on them; the excluded events are listed in Garcia et al. and reproduced in Table 1.

Excluding the seven events meaningfully reduces the flooded-residential population (202,623 $\rightarrow$ 172,329 parcels, -14.9%) and the eligible population (141,253 $\rightarrow$ 129,685 parcels, -8.2% , attributable largely to TS Allison, the only declared event among the seven). However, only 78 of the 3,432 funded parcels in the baseline are tied exclusively to the excluded events, so the funded population shifts only marginally (3,432 $\rightarrow$ 3,354 parcels, -2.3%).

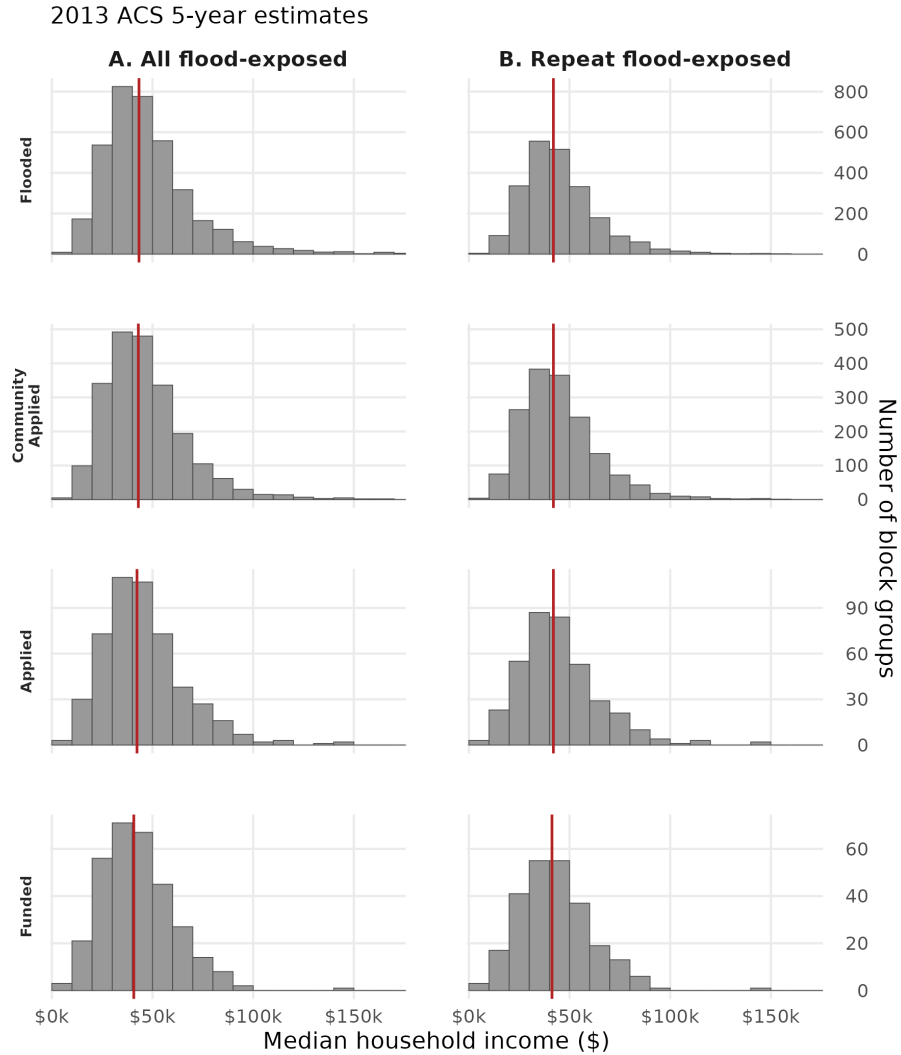


Figure 7: **Block-group median household income across the HMA pipeline.** Distribution of block-group median household income (2013 ACS 5-year estimates) for block groups containing at least one residential parcel reaching each pipeline stage. Panel A shows all flood-exposed parcels; Panel B is restricted to repeat-flooded parcels (those exposed to more than one of the 78 modeled events). Stages from top to bottom: Flooded, Community Applied, Applied, Funded. Red vertical lines mark stage medians. The x-axis is truncated at \$175,000 for readability.

2013 ACS 5-year estimates (owner-occupied housing units)

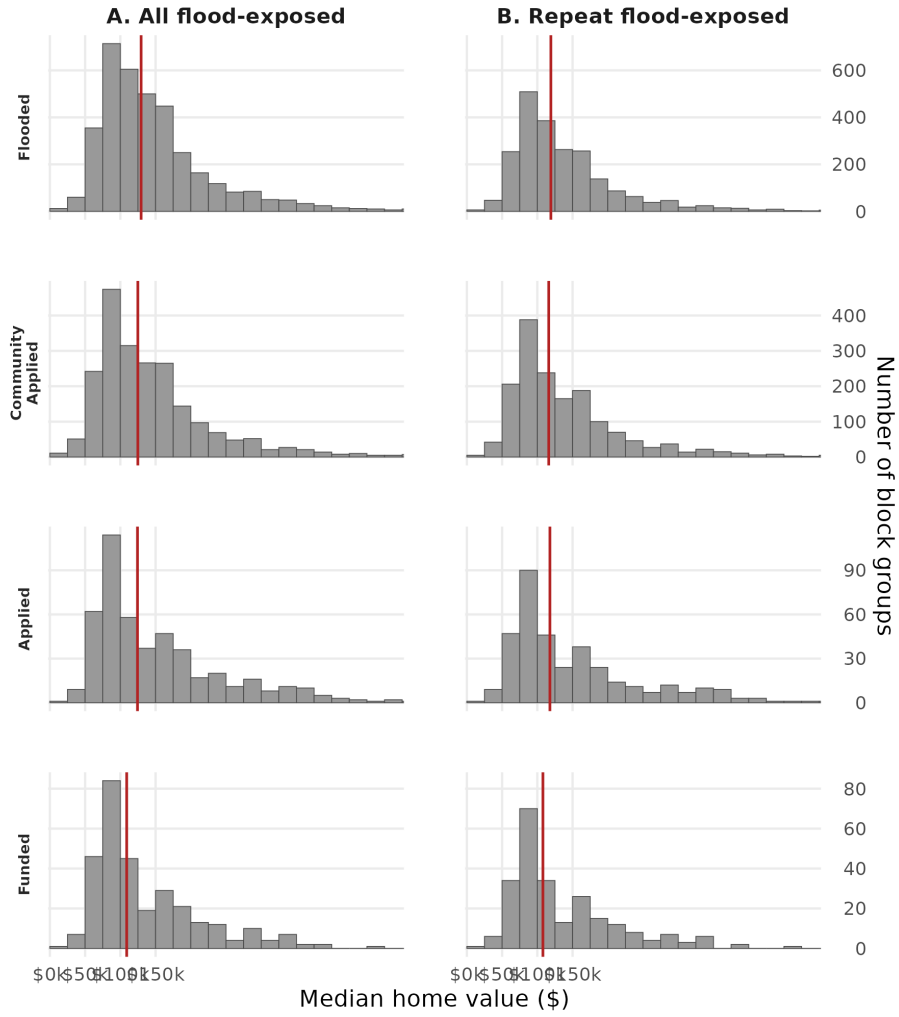


Figure 8: **Block-group median home value across the HMA pipeline.** Distribution of block-group median home value for owner-occupied housing (2013 ACS 5-year estimates) for block groups containing at least one residential parcel reaching each pipeline stage. Panel A shows all flood-exposed parcels; Panel B is restricted to repeat-flooded parcels. Stages from top to bottom: Flooded, Community Applied, Applied, Funded. Red vertical lines mark stage medians. The x-axis is truncated at \$500,000 for readability.

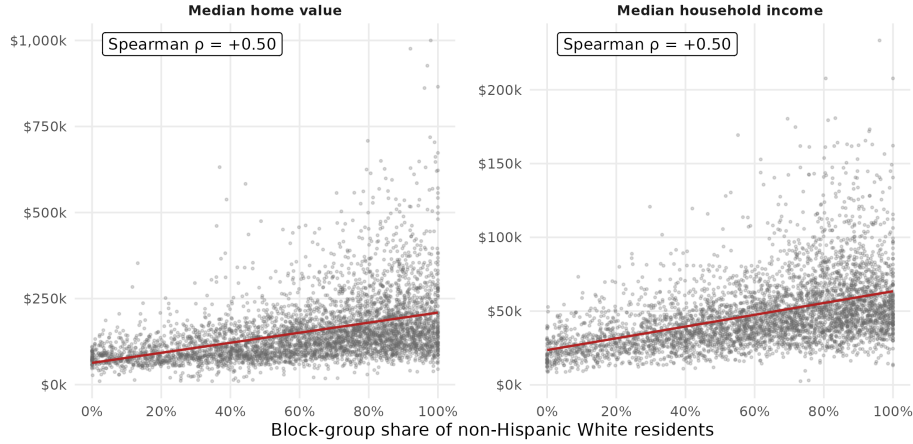


Figure 9: **Block-group correlation between race and economic indicators across the study area.** Block-group share of non-Hispanic White residents plotted against median home value (left) and median household income (right), using 2013 American Community Survey 5-year estimates for the 4,384 study-area block groups with complete data. Red lines show ordinary least squares fits; Spearman rank correlations are reported in each panel.

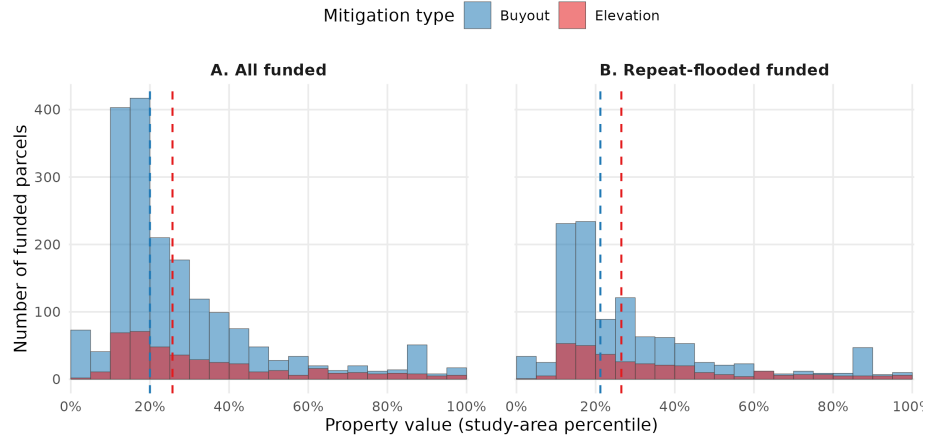


Figure 10: **Property value distributions by mitigation type, all funded versus repeat-flooded.** Study-area property-value percentile distributions for funded parcels classified as buyouts (blue) versus elevations (red). Panel A includes all funded parcels with a classified project type; Panel B restricts to repeat-flooded funded parcels. Dashed vertical lines mark within-group medians. Two-sample Kolmogorov-Smirnov tests indicate the distributions differ significantly in both panels (Panel A: $D = 0.141$, $p < 0.001$; Panel B: $D = 0.135$, $p < 0.001$).

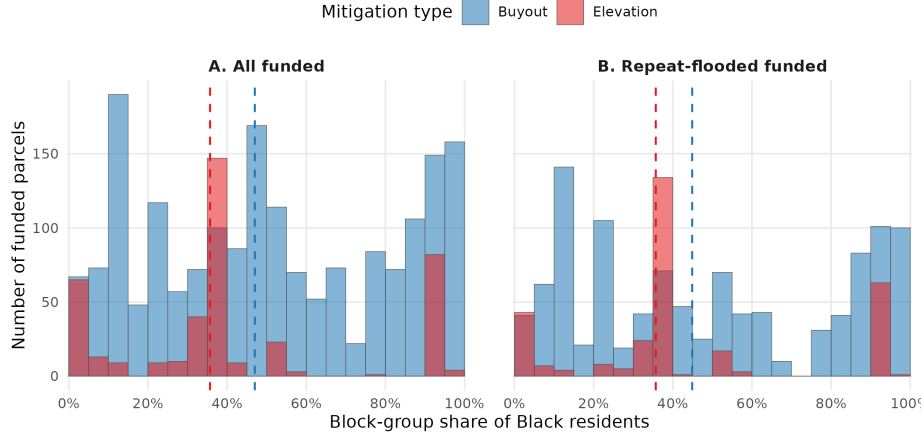


Figure 11: **Neighborhood racial composition by mitigation type, all funded versus repeat-flooded.** Block-group share of Black residents (2013 ACS) for funded parcels classified as buyouts (blue) versus elevations (red). Panel A includes all funded parcels with a classified project type; Panel B restricts to repeat-flooded funded parcels. Dashed vertical lines mark within-group medians. Kolmogorov-Smirnov tests indicate the distributions differ significantly in both panels (Panel A: $D = 0.366$, $p < 0.001$; Panel B: $D = 0.32$, $p < 0.001$), with buyouts consistently located in block groups with higher Black population shares.

The pinch point at the application stage is essentially identical across scenarios: 4.3% of properties in applying communities themselves apply under both the baseline and the 71-event sensitivity. Median funded-stage demographics and property values shift by less than half a percentage point: median NH White share moves from 60.0% to 59.6%, median Black share from 26.1% to 26.3%, and median study-area property-value percentile from the 23rd to the 23rd. Two-sample Kolmogorov-Smirnov tests of flooded-versus-funded block-group racial distributions remain highly significant under the sensitivity scenario, with D-statistics slightly larger than the baseline (NH White: $D = 0.146$ versus 0.131, $p < 0.001$; Black: $D = 0.162$ versus 0.151, $p < 0.001$). The headline pinch-point and demographic-composition findings are therefore robust to the exclusion of the seven high-uncertainty events.

Table 1: Pipeline funnel under baseline (all 78 events) and sensitivity (71 events, excluding the seven events Garcia et al. flag as having unusually high uncertainty).

Stage	Baseline (78 events)	Sensitivity (71 events)	Change
Flooded	202,623	172,329	−14.9%
Eligible	141,253	129,685	−8.2%
Community Applied	132,828	121,713	−8.4%
Applied	5,746	5,259	−8.5%
Funded	3,432	3,354	−2.3%

Table 2: Funded-stage demographics and statistical tests under baseline vs sensitivity. All KS test p-values < 0.001 in both scenarios.

Metric	Baseline	Sensitivity
Median property value (study-area percentile)	23rd	23rd
Median NH White block-group share	60.0%	59.6%
Median Black block-group share	26.1%	26.3%
KS test: NH White (D-statistic)	0.131	0.146
KS test: Black (D-statistic)	0.151	0.162